Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
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Predicting Olympic Medals Based on Enhanced Ordered-Graph Model

Summary

The Olympic Games captivate global attention, with medal count predictions often based on internal Olympic factors such as athlete performance, host country effects, and historical standings, rather than external elements like economic data. This paper develops a medal count model for the 2028 Olympics, leveraging historical data and available information to uncover key influences on medal outcomes, extract insights, and offer actionable recommendations to national Olympic committees.

Firstly, we conducted data preprocessing on the provided dataset. Due to the low dimensionality of the data, we employed **information enhancement techniques** commonly used in machine learning, predicting athletes' ethnicity based on their names and assessing national teams' adaptability to the host country's natural geography using nationality data.

Subsequently, we constructed **a Block-Slice structure**, placing each country-year data into a slice. Within each slice, we used a **graph structure** to store internal team information, where each node represents a sport, containing information about the sports team, and each edge represents the relationship between sports. We determined edge weight relationships using contrastive learning methods. We stored national feature information, which **distinguishes host and geographical factors, using** $\boldsymbol{info}$. Algorithmically, we first transformed the input graph into a matrix using graph-based algorithms. Then, we fed the matrix into our **improved Gate Recurrent Unit** for training, outputting a distribution described by a histogram to predict medals for the corresponding year. Our **Enhanced Ordered-Graph Prediction Model** performed significantly better compared to methods like ARIMA, MLR, and LSTM. Our results indicate a slight increase in medal counts for the United States, China, and the United Kingdom, a slight decrease for the Netherlands and Italy, a more significant decrease in gold medals than the increase in silver and bronze for France, and a potential breakthrough from zero to one medal for countries like Dominica, Saint Kitts, and San Marino.

Additionally, we gained further insights from the model. We discovered a **"Simpson's Paradox"** between the number of events a country participates in and its medal count, concluding that participating in more events does not necessarily increase medal counts. We captured the **"Great Coach Effect"**, proving and quantifying the significant impact of excellent coaches on team sports. We also examined the effect of doping on medals, finding that while doping may increase current medal counts, it severely damages athletes' future careers.

Our model created a data structure better suited to the data's characteristics and designed and improved a **deep learning model** with superior performance and prediction accuracy. It also captures the relationships between sports and the heterogeneity between countries, offering better interpretability compared to traditional models. However, due to deep learning's inherent limitations, its "black box" nature may restrict its application in scenarios requiring transparency and interpretability. Additionally, the high data requirements may pose significant challenges in data cleaning when faced with dirty data.

Keywords: Graph Algorithm Gated System Olympics Deep Learning Medal Prediction

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1 Introduction

1.1 Problem Background

"The Olympic Games are not just ordinary world championships but a quadrennial festival of universal youth celebrated by each succeeding generation."

— Pierre de Coubertin, the father of the modern Olympic Games

The Olympic Games, as one of the world's most significant sporting events, originated in ancient Greece and were revived in their modern form in 1896, becoming a global spectacle held every four years. It serves not only as the pinnacle of athletic competition but also as a crucial platform for showcasing culture, technology, and international cooperation, exerting a profound impact on the host country's economy, cultural exchange, and global image. Traditional sporting powerhouses, such as the United States and China, have long dominated the medal tables, while emerging forces like Japan and Australia have begun to make their mark, and the breakthroughs of smaller countries highlight the universality and diversity of the Olympics.

Today, the influence of the Olympics extends far beyond the realm of sports, becoming a focal point of research in fields such as economics, psychology, and statistics. However, most studies have predominantly relied on external factors, such as national economic strength or changes in athletes' psychology, without fully tapping into the potential of over a century's worth of Olympic historical data.(Bernard & Busse, 2004; Hedgcock, 2021) This event provides us with an excellent opportunity to let historical data "speak for itself," filling this research gap and advancing a new exploration into the prediction of Olympic outcomes.

1.2 Restatement of the Problem

Through in-depth analysis and research on the background of the problem, combined with topic-specific constraints and requirements given, the restatement of the problem can be expressed as follows:

- **Develop a predictive model for Olympic medal counts**. Build a model to predict medal counts for each country, which can provide prediction intervals of medal outcomes for the 2028 Los Angeles Olympics. Then, evaluate the model's precision and performance. Specifically, we must identify countries likely to improve, perform worse, or win their first medals.
- **Analyze the influence of Olympic events**. Explore the relationship between the events (number and types) and the medal counts for countries. Identify critical sports for various countries and assess how the host country's event selection impacts results.
- **Investigate the "great coach" effect**. Examine evidence of the impact of exceptional coaches on medal counts. Quantify the effect and propose sports where three countries should consider investing in elite coaches, estimating the potential impact.
- **Provide additional original insights**. Use the model to reveal other unique insights about Olympic medal distributions and explain their implications for country Olympic committees.

1.3 Our Work

The problem requires us to develop a predictive model that forecasts the number of medals each country will win at the 2028 Olympics, utilizing historical Olympic data and the available information about the upcoming 2028 Games. Our work mainly includes the following:

- **Organize and enhance historical Olympic data.** Using a provided explanatory table and four data tables, integrate them into a comprehensive and consistent table. Perform data mining enhancements on athlete names and exogenous features of countries (like locations) for subsequent modeling purposes.
- **Transform the table data structure into an enhanced ordered-graph data structure.** This step establishes connections between sports (via graph edges), allows for comparisons of fundamental differences between different countries and within the same country over different years, and includes the mined new information.
- **Develop an algorithm capable of unraveling the aforementioned graph data.** Utilize TG-GRU, a modified LSTM model—to learn from historical data and ultimately predict the medal standings for the 2028 Olympics, showcasing trend changes.
- **Validating the model's efficacy.** Demonstrate the high accuracy and superior performance of our strategy through longitudinal comparisons over different times and cross-sectional comparisons with different model control groups.
- **Extract additional insights from the model results and data.** Combine the model and data to provide insights into event selection, great coaching, and anti-doping measures.

To avoid complicated descriptions and intuitively reflect our work process, the flow chart is shown in Figure 1:

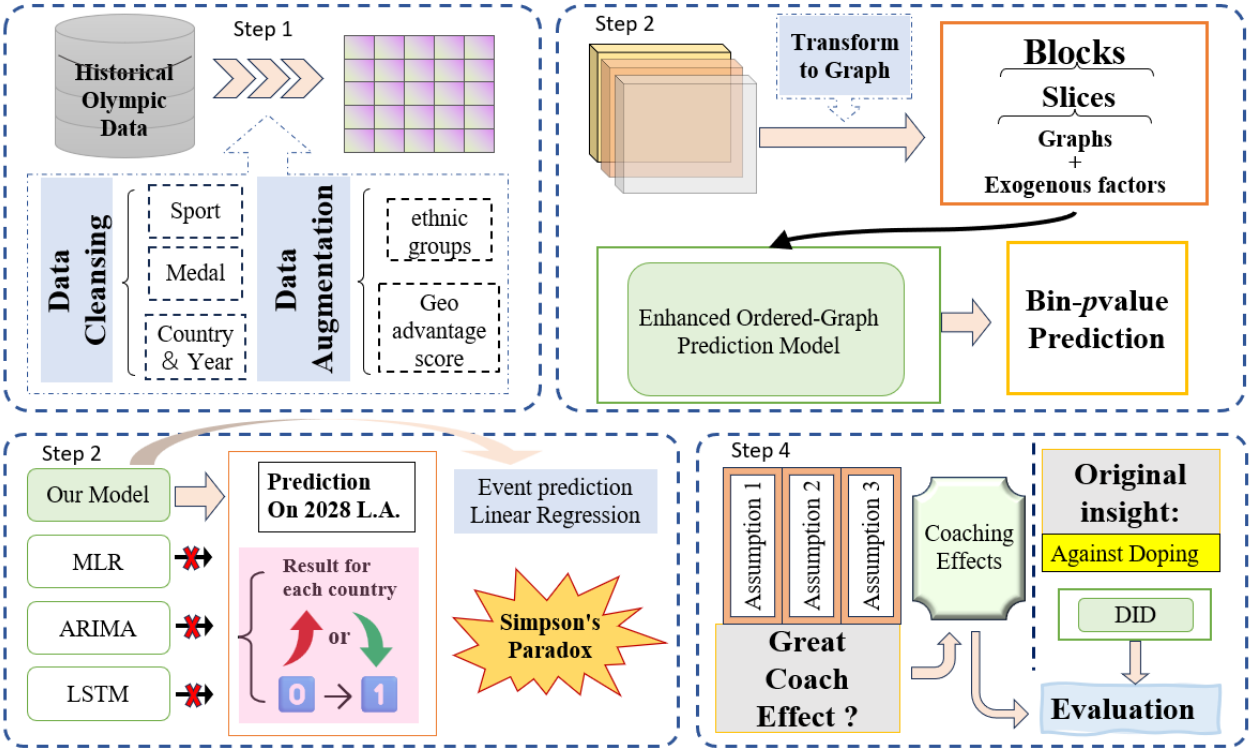

Figure 1: Flow Chart of Our Work

2 Assumptions and Notations

2.1 Assumptions

In order to simplify the model, we make the following assumptions and justify their use:

- **Assumption 1:** Despite the frequent occurrences of country divisions in reality, our model assumes that the current country's sports performance is not directly linked to that of its predecessor.
  →**Explanation:** Countries experiencing disintegration face major political and economic shifts, altering sports performance factors and making them unsuitable as continuous variables in the model.
- **Assumption 2:** Events across different sports are independent, yet within a single sport, events are interconnected. Additionally, there are correlations among sports that belong to the same overarching category.
  →**Explanation:** In aquatic sports, there is a connection between mixed relay swimming and the men's 100-meter freestyle. Similarly, baseball and cricket are distinct sports, but their techniques and strategies are closely related.
- **Assumption 3:** The athlete's name encapsulates numerous individual-level characteristics.
  →**Explanation:** For instance, an individual named Alice is highly likely to be female, while someone named José is predominantly associated with being Latino.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

2.2 Notations

Table 1 shows this paper's necessary notations and signs (***i*** represents year, ***j*** represents country). Other notations and signs will be declared or defined when using.

Table 1: Notions and Symbol Description

Symbols	Description
$mono_{ijkm}$	A country's dominance in a particular sport (*m=gold, silver, bronze, and total*)
$host_i$	The home-field advantage of the host country
geo_{ij}	The terrain, climate, and proximity to the host country
$grap_{ij}$	A repository for storing data on sports and individual athletes
$Info_{ij}$	$Info_{ij} = f(geo_{ij}, host_i)$
$slide_{ij}$	$slide_{ij} = grap_{ij} + info_{ij}$

3 Data Processing

3.1 Data Preview

Before initiating the data cleaning process, we conducted a preliminary review of the data and observed the following:

1) In *summerOly_programs.csv*, the symbol "•" denotes demonstration sports, which are not included in the Olympic medal tally.

2) *SummerOly_programs.csv* contains **51** sports, whereas *summerOly_athletes.csv* lists **76** programs.

3) In each Olympic Games, the number of medals awarded exceeds the number of events held that year (for instance, the 2008 Olympics featured **302** events, yet resulted in **303** silver, and **353** bronze medals).

3.2 Data Cleansing

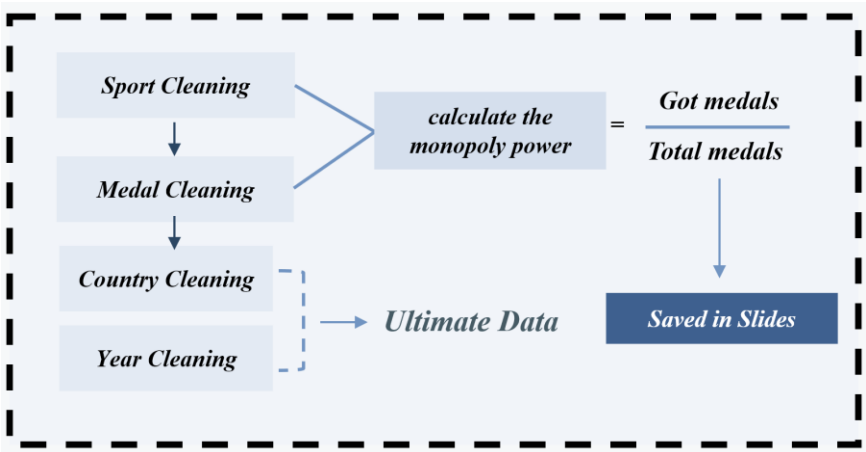

Figure 2: Flow Chart of Data Cleansing

3.2.1 Sport Cleaning

We started by cleaning *summerOly_programs.csv*:

1) Remove demonstration sports.

2) Incorporate sports, principles, and events for the 2028 Olympics.[1]

3) Exclude newly introduced sports for 2028 (due to the lack of historical data, predictions are not feasible, but estimations can be made using analogous sports).[2]

4)Eliminate all sports not featured in the 2028 Olympics.

[1] Based on the committee's amendments to this topic, we are permitted to query data related to the 2028 competition events. The information provided in this response regarding the 2028 Olympic Games is sourced from https://en.wikipedia.org/wiki/2028_Summer_Olympics and https://la28.org.

[2] Cricket was held once before in 1900, but has not been held since until 2028, so it is treated as if it had never been held before.

After completing this part of the cleaning, **31** valid sports were left behind. Subsequently, we harmonize the 'Sport' column in both *summerOly_programs.csv* and *summerOly_athletes.csv*:

1) Complete the 'event' entries in *summerOly_athletes.csv* to prevent any loss of original data.

2) Systematically replace the 'Sport' in *summerOly_athletes.csv* with the corresponding entries from the program.

3.2.2 Medal Cleaning

The essence of team competitions is that multiple individuals collectively earn a portion of a medal for their country (e.g., in a four-person team event, each contributes 0.25 of a gold medal). Therefore, we proceed as follows:

1) First, transform the Medal1 column into four columns, using 0 and 1 to indicate whether the corresponding medal was won (or not).

2) Reorder the summerOly_athletes.csv file based on 'Year', 'Sport', 'Event', and 'NOC'.

3) Identify hierarchical relationships; if all attributes except gender are identical, classify it as a team event, with each individual credited with 1/n of the medal.

Before conducting a country cleansing, we need to calculate **the monopoly power**:

$\boldsymbol{mono_{ijk}}$**:** the number of medals a country $\boldsymbol{j}$ can win in year $\boldsymbol{i}$ for sport $\boldsymbol{k}$

This $mono_{ijk}$ is then incorporated into the subsequent graph. The rationale behind this approach is that the total number of gold medals awarded in each sport **fluctuates**, necessitating the use of **relative values rather than absolute ones** for accurate measurement.

3.2.3 Country and Year Cleaning

According to **Assumption 1** stated earlier, we will exclude countries not under consideration. Additionally, we will remove data from the following three years:

- 1896: The first Olympic Games, primarily featuring amateur athletes, with only male participants and a limited number of competitors
- 1906: The status of this event as an official Olympic Games is disputed
- 1980: Boycotted by Western countries, resulting in too few participating countries after excluding non-existent countries.

3.3 Data Augmentation

Athletes from various countries travel to the Olympic host city to compete, but due to the limitations of the Olympic Village, they often arrive and depart in staggered groups. The fatigue from travel or acclimatization issues may prevent them from reaching their optimal condition before the competition begins. The varying distances between different countries and the host city and differences in latitude, longitude, and climate can impact the entire team. Therefore, we assess the geographical advantage score: geo_{ij}, by evaluating the distance and environmental similarity between the athletes' home countries and the host city.

$$geo_{ij} = F(topographic, climate, distance)$$

Ethnic advantages play a crucial role in various sports, as different ethnic groups excel in distinct events. Consequently, we predict athletes' ethnicities using their names and nationality data, determining the probability of each athlete belonging to specific ethnic groups. The enhancement of these two data types is illustrated in Figure 3.

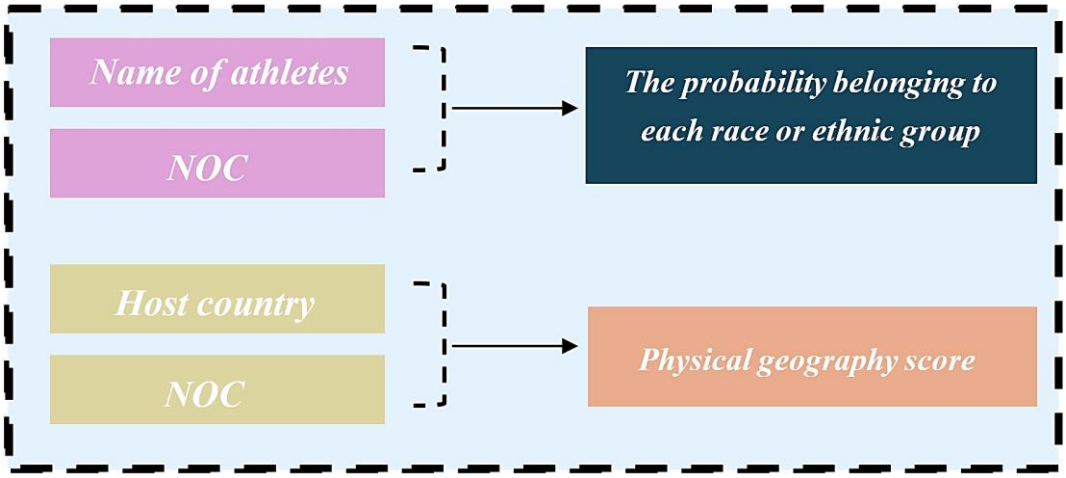

Figure 3: Feature detection and data Augmentation

3.4 Descriptive Statistics

Now that we have obtained the final dataset, we can perform descriptive statistics on it, as illustrated in Figure 4 and Table 2 :

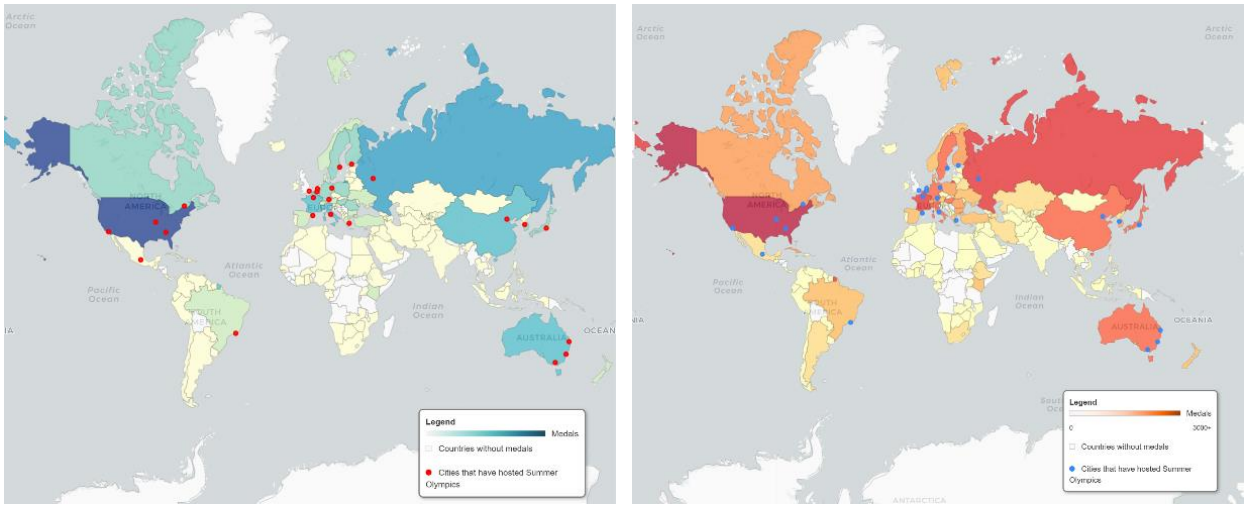

Figure 4: Global Medal Distribution Since the 2004 Olympics

(Left: Total Medal Distribution, Right: Gold Medal Distribution)

Table 2: Descriptive Statistics

	count	unique	top	freq	mean	std	25%	50%	75%
Name	220340	114537	John Jr.	74					
Sex	220340	2	M	153099					
NOC	220340	206	USA	15900					
Year	220340				1984.785	31.98657	1964	1992	2012
City	220340	22	London	21276					
Sport	220340	31	Athletics	38564					
Event	220340	958	Football Men's Football	4904					
Gold	220340				0.021747	0.131148	0	0	0
Silver	220340				0.021711	0.131115	0	0	0
Bronze	220340				0.024089	0.139296	0	0	0
No medal	220340				0.85679	0.350288	1	1	1

4 Enhanced Ordered-Graph Prediction Model

Our comprehensive model integrates an enhanced ordered-graph data structure (we proudly call it **Slice**, which looks like a slice of bread) with the Triple-Gated Gate Recurrent Unit (**TG-GRU**) data algorithm improved from LSTM, ultimately forecasting the medal count for 2028. Figure 5 illustrates the detailed implementation process of the entire model.

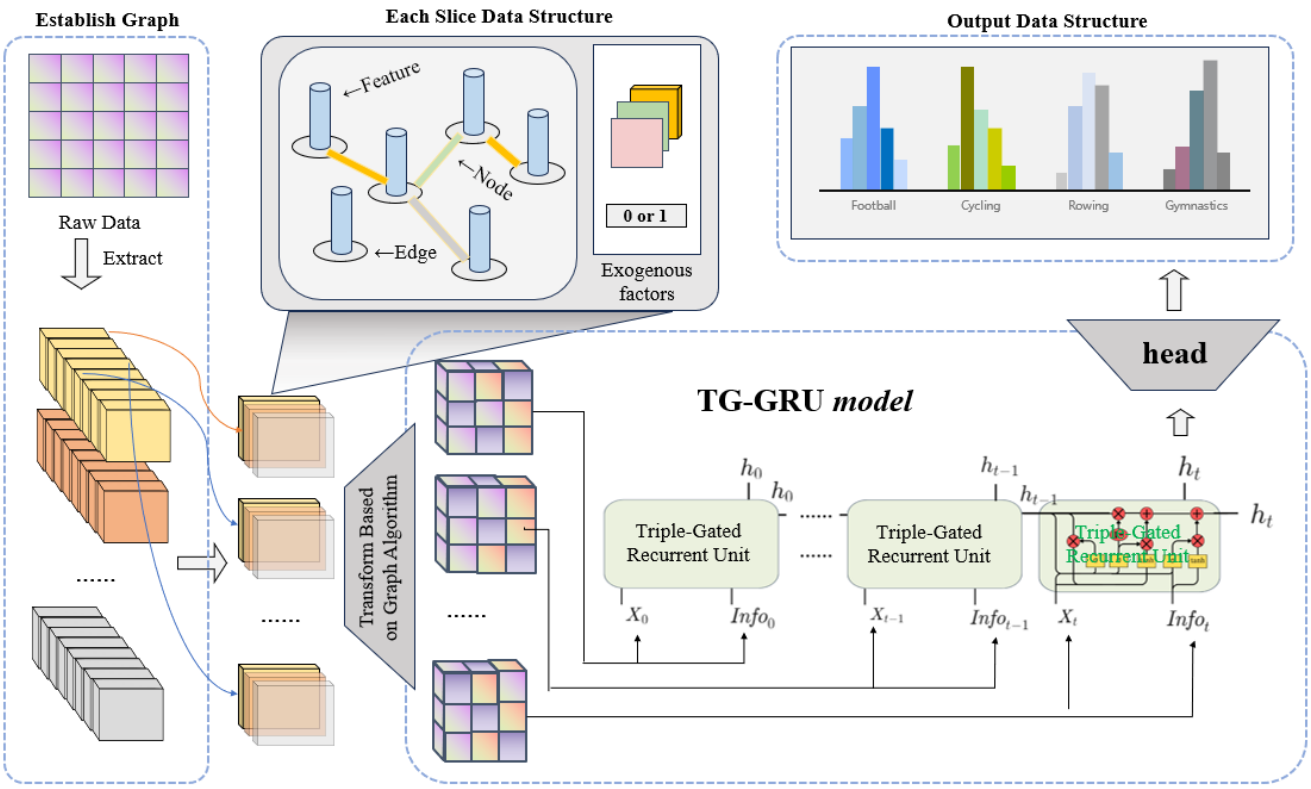

Figure 5: Total Precession of Enhanced Ordered-Graph Prediction Model

4.1 Data Structures: Block-Slice

4.1.1 Establishment of Graph Structures

We categorized the data of all athletes and events into 206 **blocks** based on different countries. Subsequently, these blocks were organized into 35 **Slices** according to the chronological order of the Olympic Games. Each Slice represents the complete information of a country's participation in a specific Olympic Games, containing **graph (node + edge)** and **attached information**. Within each Slice, there are covariates describing the characteristics of athletes and events, as well as **label** variables indicating the outcomes of the Olympic Games. The aforementioned process is illustrated in Figure 6.

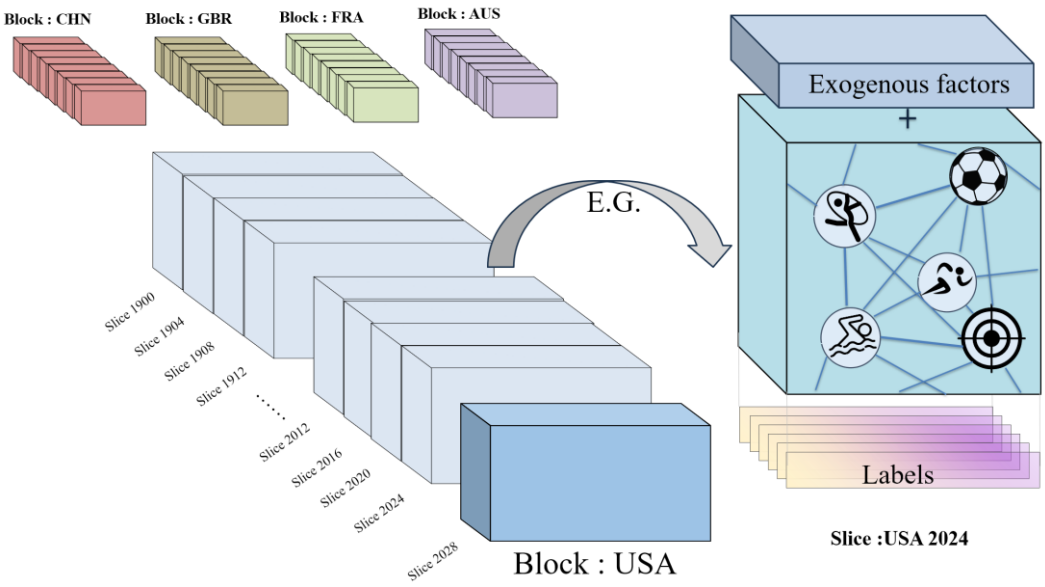

Figure 6: How We Build up Our Data Structure

4.1.2 Establishment of Nodes and Edges

We consider each sport as a **node**, resulting in 31 nodes for 31 sports. Each node contains information about the athletes and the number of events for that particular sport (Figure 7).

Participated event numbers represent how many specific events a country participated in during a given year, indicating the potential medal opportunities based on the number of events entered.

All event number refers to the total number of events for this sport in the current Olympic Games, which corresponds to the total number of gold medals available.

Gender statistics provide a count of male and female athletes in the teams for the current Olympics.

The probability prediction of ethnic groups, derived from the analysis of names and nationalities, categorizes athletes into Asian, African, and European races. By summing the racial probabilities of all team members, we obtain the expected number of individuals from each ethnic group within the team.

Veteran athletes with extensive Olympic experience can enhance the team's morale, and we have statistically determined the number of such veterans in the team.

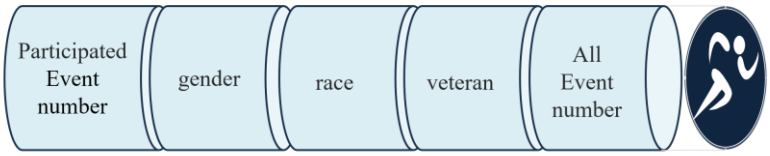

Figure 7: What Are Contained in a Node

Edges in a graph represent the relationships between nodes, which in our model correspond to the connections between sports, representing the relationship between each two sports.[3] Meanwhile, by comparing the graphs among different blocks, we can obtain the heterogeneity among countries. This relationship can be derived using contrastive learning methods, as illustrated in Figure 8.

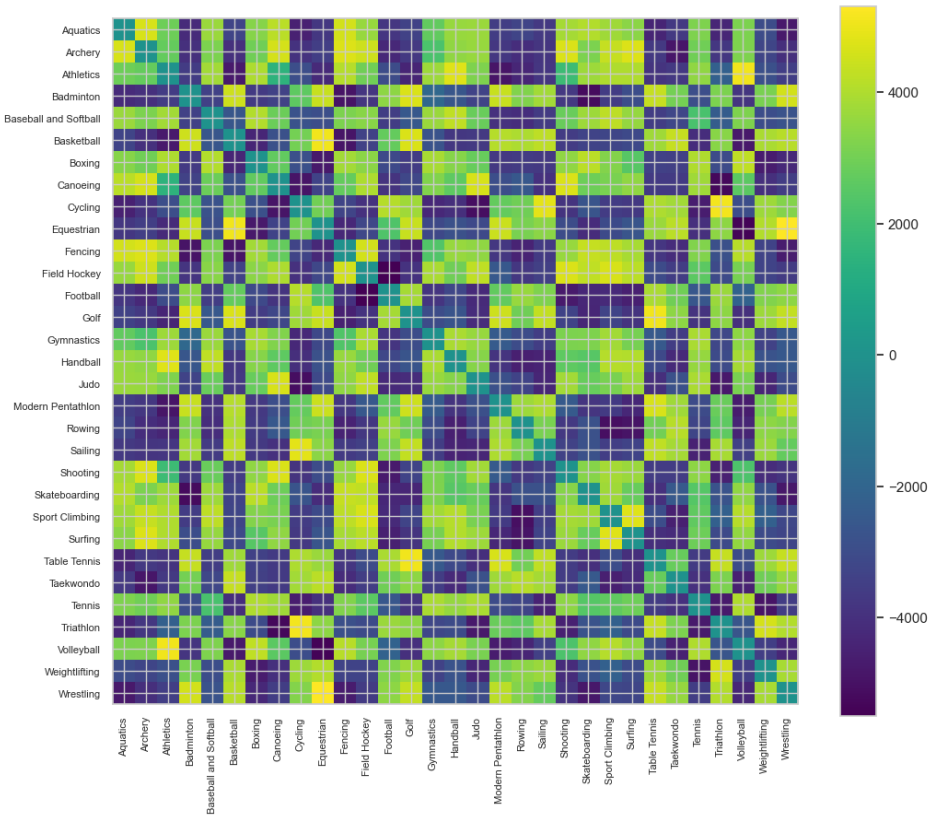

Figure 8: The Relationships Between Sports as Represented by Edge

4.1.3 Attached Information

Each graph has attached information that characterizes a specific county in that year: the host country enjoys a range of home-field advantages, such as referee bias and fan support. Thus, the attached information includes the host status as a dummy variable. The geographical advantage score reflects the travel fatigue and adaptability to local altitude and climate, which can impact the team's performance.

$$Info_{ij} = f(geo_{ij}, host_i)$$

[3] See the Appendix for a detailed implementation of this section

4.1.4 Constructing Labels

We compare the medal tally of our national team in a specific sport, including gold, silver, bronze, and total medals, with the total number of medals awarded in the corresponding sport at the current Olympic Games, using this as the data label. The proportion of gold medals won relative to the total number of events reflects the country's dominance in securing gold in that sport, and the same applies to silver, bronze, and total medals. For instance, if the Chinese table tennis team wins all the gold medals in 2008 Olympic Games, then China's gold medal dominance label in table tennis would be 1.

$$mono_{2008\ CHN\ table-tennis\ gold} = 1$$

4.2 Algorithm Design: Triple-Gated Gate Recurrent Unit (TG-GRU)

Traditional LSTM algorithms have significantly mitigated the forgetting problem inherent in RNNs. However, they come with a complex structure, slow training speeds, and potential incompatibility with our data structure. Consequently, we have innovatively modified the conventional recurrent gating units to better align with our data structure. Below is a detailed explanation.

4.2.1 Transform Based on Graph Algorithm

When processing the data from our graph structure, we employ a transform based on graph algorithm to extract valid information from the graph. This includes node features representing different sports, edge features depicting relationships between sports, and the structural information of the entire graph. By aggregating the information from the entire graph, the transform based on graph algorithm can generate a global representation of the graph, which serves as the input matrix for the subsequent step.

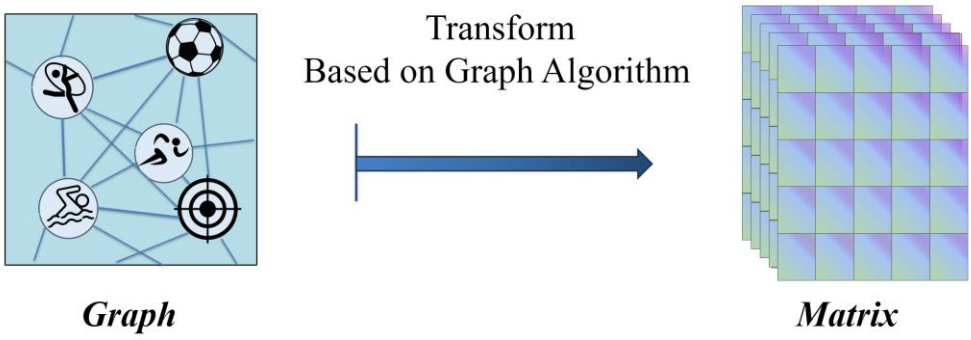

Figure 9: Transform Based on Graph Algorithm

4.2.2 Triple-Gated Gate Recurrent Unit

Each unit is divided into three gates: the update gate, the reset gate, and the reinforcement gate.

- **Update gate:** It determines the extent to which historical data should be retained up to the present moment and the degree to which current input information should be integrated into the new hidden state.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$$

In this equation, W_z and b_z represent the parameter matrix and bias vector, respectively. h_{t-1} denotes the hidden state from the previous time step, while x_t signifies

the current input. The function $\boldsymbol{\sigma}$ refers to the activation function. When z_t approaches 1, it indicates a greater retention of the past hidden state; conversely, when z_t nears 0, it suggests a stronger reliance on the current input to update the hidden state.

- **Reset gate:** It determines to what extent the hidden state of the previous time step is ignored.

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$$

When the output of the reset gate approaches **0**, the network tends to "forget" the information from the previous time step, relying solely on the current input. Conversely, when the output nears **1**, the information from the previous time step is more likely to be retained.

- **Reinforced gate:** We introduce the reinforcement gate, which aims to allow the model to utilize $Info_t$ of the Slice as y_t to dynamically adjust the update of the hidden state, thus improving the model's ability to represent complex tasks. By adjusting the weights through the reinforcement gate, the model can explicitly incorporate the effect of auxiliary information on the prediction results based on the first two gates.

$$s_t = \sigma(W_s \cdot y_t + b_s)$$

The output value of the reinforcement gate ranges between 0 and 1, with higher values indicating a stronger influence of y_t on the hidden state.

- **Ultimately**, the computation method for h_t is as follows:

$$h_t = (1 - z_t) \cdot h_{t-1} + \widetilde{h_t^1} \cdot z_t + \widetilde{h_t^2} \cdot s_t$$

The coefficient in front of h_{t-1} is reversed using the method $(1 - z_t)$, which gives the trained z_t **the practical interpretation of a forgetting rate**. A larger z_t value results in a smaller weight for h_{t-1}, implying that more data is updated. This establishes a positive correlation between the magnitude of z_t and the extent of updates, enhancing its practical significance. Consequently, after inputting x_t and y_t in each iteration and applying the aforementioned transformation, supervised learning with labels transforms h_{t-1} into h_t.The complete process is shown in Figure 10.

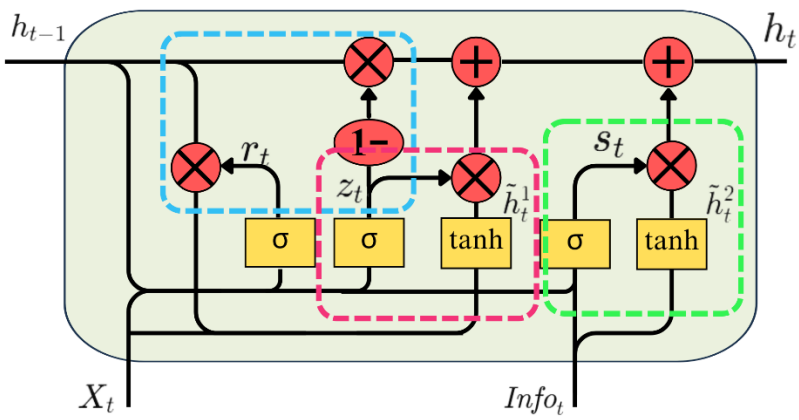

Figure 10: Triple-Gated Gate Recurrent Unit

4.2.3 From Units to Layer

Starting with data from the 1900 Olympics, select all Slices from each Olympic Games, label them with the respective country, and input them into the model. The data from each Olympic Games is used as x and y and fed into the TG-GRU, updating h_{t-1} to h_t. The h_t serves as the output to the HEAD and is also retained in the system as ht-1 for the next unit.

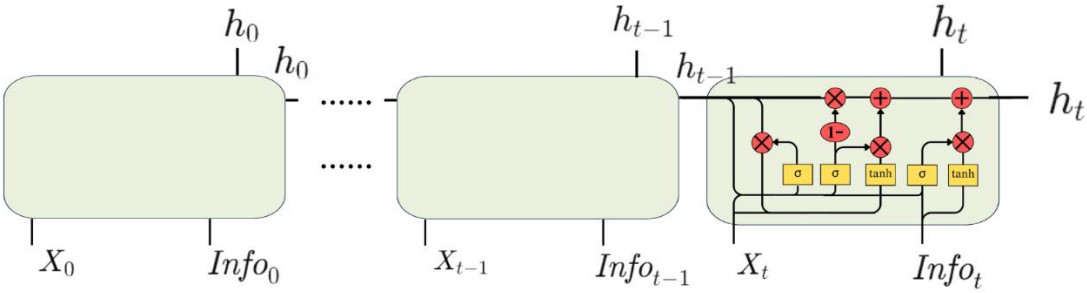

Figure 11: Triple-Gated Gate Recurrent Unit Layer

4.2.2 Data Output and Estimation

In this step, to transform each step's intermediate output h_t into an intuitive probability distribution, we designed a HEAD mechanism. Its primary goal is to map the continuous value h_t to a set of predefined interval probabilities.

Specifically, we first divide the interval [0,1] into 10 equal sub-intervals: [0,0.1], [0.1,0.2], [0.2,0.3], and so on, up to [0.9,1.0]. Next, a fully connected layer is applied to project h_t into a distribution that matches these intervals. The computation of the fully connected layer is defined by the following formula:

$$P = Softmax(W \cdot h_t + b)$$

Here, W is the weight matrix, b is the bias vector, and $Softmax$ is a commonly used activation function that normalizes the output into a probability distribution. This ensures that the probability corresponding to each interval lies between 0 and 1, and the sum of all interval probabilities equals 1.

With this design, the HEAD module converts the intermediate result h_t into a more intuitive and practically meaningful representation, i.e., the probability distribution across different intervals. This provides clearer insights for subsequent analysis or classification tasks.

During the model training process, it is noteworthy that we employed the cross-entropy loss function for optimization:

$$Loss = a\sum_i\sum_j - y_i \log\left(P\left(y = label_j|x\right)\right) + b\sum_i \sqrt{\frac{1}{n}\sum_j \left(y_i - label_j\right)^2}$$

Maximize the likelihood estimation for the predicted probabilities of each bin to ensure the accuracy of the estimation.

To forecast for 2028, we trained using all previous x_t data and employed an ARIMA model to predict the x_t for 2028, which serves as the input for that year. The y_t is known since the event

location has already been announced. These are fed into the model to obtain the result data for 2028. The code implementation for the entire process is presented as follows:

Algorithm 1: Enhanced Ordered-Graph Prediction Model

```
Data: Node features V_t, adjacency matrix E_t, additional features Y_t,
      labels L_t
Result: Trained model parameters θ
Initialize: Model parameters θ, learning rate η, hidden state
  h_prev ← 0, number of epochs N, and optimizer;
for epoch ← 1 to N do
    TotalLoss ← 0;
    foreach year t in sample_dict do
        Extract all samples {(V_t, E_t, Y_t, L_t)} for year t;
        foreach sample (V_t, E_t, Y_t, L_t) do
            if t == 1900 then
                h_prev ← 0;
            end
            else
                h_prev ← h_pred;
            end
            Compute predictions (output_1, output_2, output_3, output_4) and
              h_pred using model M(V_t, E_t, Y_t, h_prev);
            SampleLoss ← 0;
            foreach sport s in ALL_SPORTS do
                Compute predicted gold, silver, bronze medal
                  distributions using output_1, output_2, output_3;
                Construct target distribution based on L_t and normalize
                  by total medals;
                Compute loss Loss_s for sport s using cross-entropy;
                SampleLoss ← SampleLoss + Loss_s;
            end
            Backpropagate SampleLoss and update θ using optimizer;
            h_pred ← h_pred.detach();
        end
        TotalLoss ← TotalLoss + SampleLoss;
    end
    if epoch mod 1 == 0 then
        Print TotalLoss;
    end
end
```

5 The Model Results

5.1 Basic Results: Model Precision and Performance

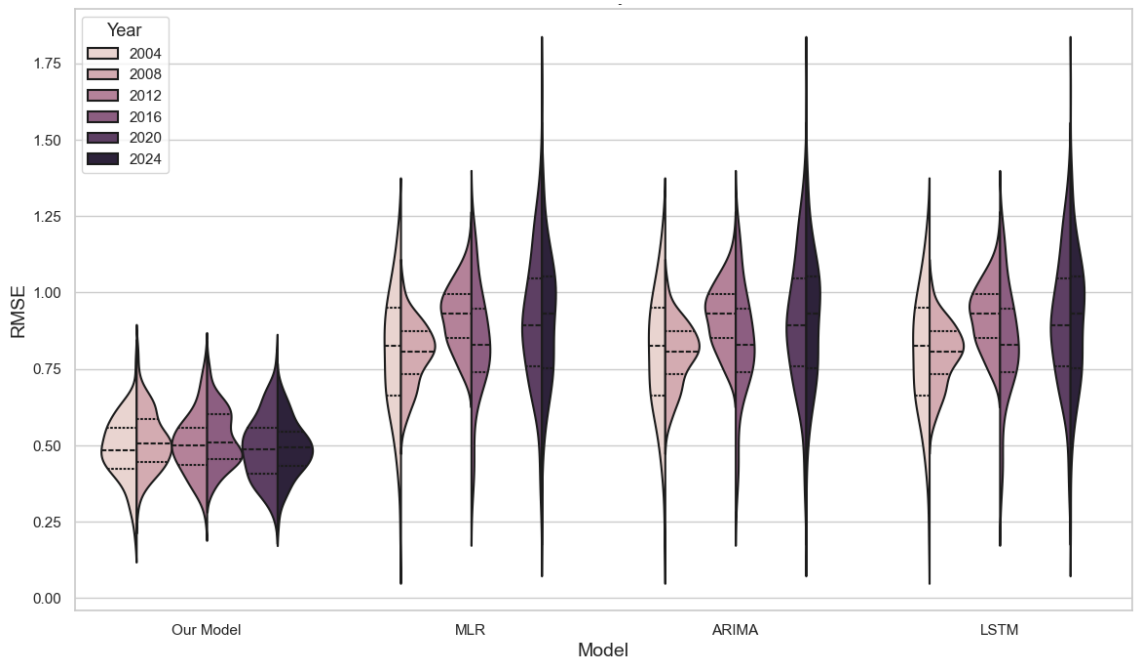

Figure 12: RMSE Distribution by Model and Year

Our model demonstrates superior performance compared to common models typically used for predicting competition outcomes. For a unified comparison, we employed the Root Mean Square Error (RMSE) as the metric for evaluating the models. We processed the slice and label data, using them as inputs (X) and outputs (y), respectively, to train multiple models, including multiple linear regression (MLR), ARIMA, and Long Short-Term Memory (LSTM) networks. The RMSE values for predictions of medal outcomes across nearly six recent Olympic Games were recorded, as shown in Figure 12.

As depicted in the figure, our model exhibits a clear advantage in predicting medal outcomes for countries over the past six Olympic Games. The figure illustrates the distribution of RMSE values for each model, with our model consistently achieving lower average RMSEs compared to the other models. This indicates a higher predictive precision for our model. Additionally, the final distribution of RMSEs for our model is the most concentrated, suggesting that our model demonstrates a high degree of robustness.

In summary, these results highlight the outstanding performance of our model, underpinned by its superior predictive precision and robustness. These findings validate the efficacy of our approach in comparison to conventional forecasting models.

5.2 Predicted Results: 2028 Olympic Games

5.2.1 Addition of New Sports

Three new sports, Cricket, Hockey and Squash, have been added to the 2028 Los Angeles Olympics. But for the reason that our model is based on historical data, these are the sports that we have no way

to put into our model to predict. However, we surprisingly found that these programs are highly consistent with the hitting programs such as Hockey and Baseball, and we also found a high correlation between Baseball and Hockey in our previous model. Therefore, for all three new programs, we can use our previous estimates for each country's access to baseball and field hockey in 2028 to estimate the new programs.

Flag football, one of the four new sports, has the same characteristics, but its rugby characteristics are weakly correlated with existing Olympic SPORTS, so it is not estimated.

5.2.2 Medal Table of 2028 Olympic Games

Here we show the model's prediction of the medal distribution for the 2028 Olympics. Due to space constraints, only the 95% confidence intervals for the number of medals for the top ten countries and their trends are shown.

Table 3: 2028 Olympics Medal Table Top 10 Predictions and Their Trend Changes

Country	Total	Gold	Silver	Bronze
USA	**124** ↑ (±-0.24)	**43** ↑ (±3.00)	**42** ↑ (± 1.74)	**39** ↑ (±1.90)
CHN	**91** ↑ (± 3.13)	**40** ↑ (±2.59)	**30** ↑ (± 1.05)	**21** ↓ (± 1.19)
GBR	**65** ↑ (± 3.86)	**15** ↑ (±1.81)	**21** ↓ (± 0.96)	**29** ↑ (±2.63)
AUS	**53** ↑ (±1.48)	**19** ↑ (±2.94)	**17** ↑ (± 0.60)	**17** ↑ (± 1.04)
FRA	**52** ↓ (± 3.45)	**22** ↓ (±3.85)	**19** ↑ (± 2.70)	**11** ↓ (± 1.30)
RUS	**31** ↑ (±3.63)	**13** ↑ (± 3.09)	**11** ↑ (±2.72)	**7** ↑ (± 3.73)
JPN	**47** ↑ (± 1.30)	**21** ↑ (± 2.79)	**12** ↓ (±0.39)	**14** ↑ (±3.20)
ITA	**42** ↓ (± 0.79)	**14** ↑ (± 1.77)	**13** ↓ (± 0.88)	**15** ↓ (± 0.14)
KOR	**37** ↑ (±1.46)	**15** ↑ (±3.85)	**10** ↑ (± 1.41)	**12** ↑ (±0.63)
NED	**32** ↓ (± 1.87)	**13** ↓ (±0.88)	**10** ↑ (± 1.89)	**9** ↑ (±2.24)

5.2.3 Breakthrough from Zero to One

The modern Olympic Games have been held for over a century, with more than 200 countries participating in each edition since the 20th century. However, some nations have yet to secure their first Olympic medal. Despite this, they are actively preparing with hopes of earning their inaugural medal at the 2028 Games. Our model offers promising predictions for such a breakthrough from "zero to one."

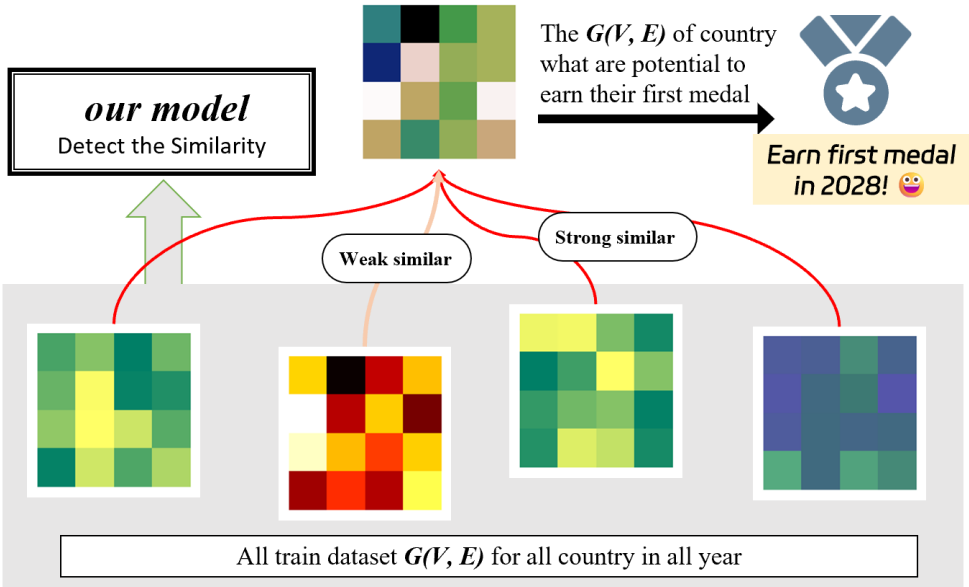

Figure 13: Principles of Predicting National Breakthroughs

As illustrated in Figure 13, the core principle behind our model's prediction of a "zero to one" breakthrough is rooted in the analysis of historical data. By training on datasets where certain countries have transitioned from winning no medals to securing their first, our model identifies key indicators that precede such achievements. These indicators include factors such as information on the composition and arrangement of the team, and previous participation records, which collectively reflect a nation's potential to achieve an Olympic medal. Through 100 iterations of model training, our approach captures patterns and trends that are indicative of this progression.

In our predictions, countries such as Dominica, Saint Kitts and Nevis, Saint Lucia, and San Marino are expected to experience their own historic breakthroughs. This can be further understood by analyzing data patterns observed in the predicted team compositions for these countries in 2028, which exhibit specific trends and features. Through the observation of these data, we found that the team lineups of these countries predicted by ARIMA in 2028 showed some regularities and characteristics. The breakthrough of Dominica, in particular, stands out which can be related to its recent investments in sports infrastructure and youth development programs, having significantly enhanced its athletes' performance.

Ultimately, our model successfully applies these insights to practical scenarios, providing significant encouragement and support for the athletic training of these nations. The predictions offer valuable insights into the potential trajectory of Olympic performance, fueling hope and motivation for countries striving for their first Olympic medal.

6 Other Results

6.1 Event vs. Medal: the Simpson Paradox in the Olympics

In the Olympic Games, nations strive not only for their standings on the medal table but also for a comprehensive development in competitive sports. To explore the relationship between the number

of unique event entries and the total number of medals, we employed a linear regression model, as represented by the following equation

$$Total_{Medals} = \beta_0 + \beta_1 \cdot Unique_Events_Count + \epsilon$$

By fitting the model, we obtained β_1=0.2245 and β_0=−2.63. This indicates that for each additional unique event entered, the total number of medals increases by an average of approximately 0.2245, indicating a positive overall correlation between event diversity and medal success. This reflects two factors: a stable medal probability with more events and the growing competitiveness of countries expanding into new sports.

However, this trend doesn't hold universally. **Simpson's Paradox** emerges as some countries show an inverse relationship between event participation and medal counts. For example, in the 2020 and 2024 Olympics, Germany, the Czech Republic, Switzerland, the Netherlands, Nigeria, and Slovenia participated in significantly more events yet secured fewer medals. This paradox highlights the impact of resource dispersion, where reallocating funding and training to new events may weaken traditional strengths, aiming for long-term gains but sacrificing immediate results.

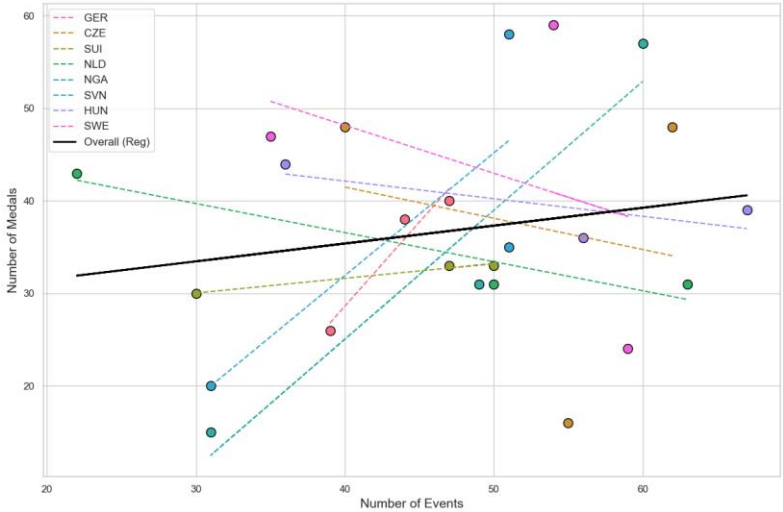

Figure 14: The Simpson Paradox in the Olympics of 10 Countries

Resource disparities amplify these effects. Powerhouses like the U.S. benefit from broader participation, whereas smaller nations like the Bahamas struggle to sustain medal counts when expanding into new events.

In conclusion, while aggregate data shows positive trends, stratified analysis reveals nuanced dynamics shaped by resource allocation and strategy. Balancing global trends with local insights is crucial for optimizing sports development and achieving Olympic success.

6.2 Great Coach Effect

To identify the coaching effect, we must first integrate the existing data.

Assumption 1: The coaching effect should result in **a significant increase** in the number of medals in a specific sport category, which could either be a breakthrough from zero to one or a substantial increase compared to previous counts.

Assumption 2: The role of a coach varies across different sports. For instance, sports like football, basketball, and volleyball, which rely heavily on **tactical arrangements**, are more likely to exhibit a star coach effect. In contrast, sports such as long-distance running, triathlon, and swimming are more dependent on individual talent.

Therefore, we merged and cleaned the data to create a medal statistics table categorized by country, year, and sport. In this table, we analyzed significant medal changes (from zero to one or at least **3** times the previous medal count, with an increase of at least **4** medals), resulting in 1998 data entries.

Focusing on sports where the coaching effect is more pronounced, we retained only football, basketball, volleyball, hockey, and handball, narrowing it down to 159 entries. By combining renowned coaches provided in the study with those identified through online sources, we identified entries where performance "leaps" were attributed to star coaches, as shown in Table 4.

Table 4: Significant Great Coach

NOC	Sport	Year	Medals	Previous Year	Previous Medals	Medal Increase	**Great Coach**
Romania	Volleyball	1980	1	1972	0	1	**Béla Károlyi**
United States	Volleyball	1984	2	1968	0	2	**Doug Beal**
Spain	Football	1992	1	1980	0	1	**Vicente Miera**
United States	Football	1996	1	1992	0	1	**Tony DiCicco**
Argentina	Basketball	2004	1	1996	0	1	**Rubén Magnano**
United States	Volleyball	2008	2	2004	0	2	**Lang Ping**
China	Volleyball	2016	1	2012	0	1	**Lang Ping**
Italy	Volleyball	2024	1	2020	0	1	**Julio Velasco**
Poland	Volleyball	2024	1	2020	0	1	**Nikola Grbić**

We find that in addition to Lang Ping and Béla Károlyi, given in the title. there are also very significant marquee effects, the closest of which is the 2024 Paris Olympics, where current Italian women's volleyball coach Julio Velasco, who has led the Italian men's team to consecutive World Championships and World Cups since the mid-1990s, is once again leading the Italian national team to the Olympic title .

Assumption 3: For a sports team, there is a *coaching effect* and an *athlete effect*, and we assume that coaching will only bring about tactical changes and cannot change the characteristics of the athletes themselves.

$$Coaching_effect + Athlete_effect = 1$$

Therefore, we can compare the team that won the title under a great coach with the team that didn't perform as well in the last Games. In addition to coaching changes, there were also changes in the **athlete ranks** in both Olympics, and we can measure the athlete effect by **the athlete turnover rate**,

so the remaining part of the equation is the coaching effect. In Table 5 below we list a couple of coaching effects for different teams.

Table 4: Coaching Effects Analyzed from Total Effects

	Total number of medal-winning years	Those who were there in the last Games	turnover rate	**coaching effect**
Poland Men's Volleyball 2024	13	9	0.308	**0.692**
Italy Women's Volleyball 2024	13	6	0.538	**0.462**
China Women's Volleyball 2016	12	3	0.75	**0.25**
U.S. Women's Volleyball Team 2008	12	7	0.417	**0.583**

It can be found that the coaching effect is **different** for different countries, different genders and different programs, and a uniform answer cannot be given. But from the data I found, volleyball is a very famous coaching effect of the sport, in addition to the two examples given in the question, multiple cases from volleyball. I found that volleyball teams from South Korea, Poland, and Canada have made it to the top eight or top sixteen of the Olympic volleyball tournament several times but have yet to win a medal, so I recommend hiring **a great volleyball coach** for these three countries.

6.3 Doping: Killing the Goose that Lays the Golden Eggs

Doping has long been a contentious issue in competitive sports. While it can significantly enhance athletes' performance in the short term, it undermines the fairness of competition and is strictly prohibited. However, the long-term effects of doping remain controversial. Does doping represent a “killing the goose that lays the golden eggs” strategy, boosting short-term performance at the expense of long-term outcomes? To investigate this question, I employ panel data on annual medal counts from international sports competitions and utilize a Difference-in-Differences (DID) framework to estimate the causal effects of doping on both short- and long-term medal outcomes.

The model is specified as follows:

$$Total_Medals_{it} = \beta_0 + \beta_1 \cdot Doping_Year_{it} + \beta_2 \cdot Post_Doping_{it} + \beta_3 \cdot (Doping_Year_{it} \times Post_Doping_{it}) + \epsilon_{it}$$

The DID estimation reveals that doping has significant short- and long-term effects on medal counts. In the year of doping, medal counts increased by **3.21** ($p < 0.001$) relative to non-doping years. However, in subsequent years, medal counts significantly declined, with a decrease of **4.59** ($p < 0.001$). The interaction term indicates that the short-term gains from doping are offset by a persistent downward trend in the long term, consistent with the hypothesis of a “scorched-earth” effect.

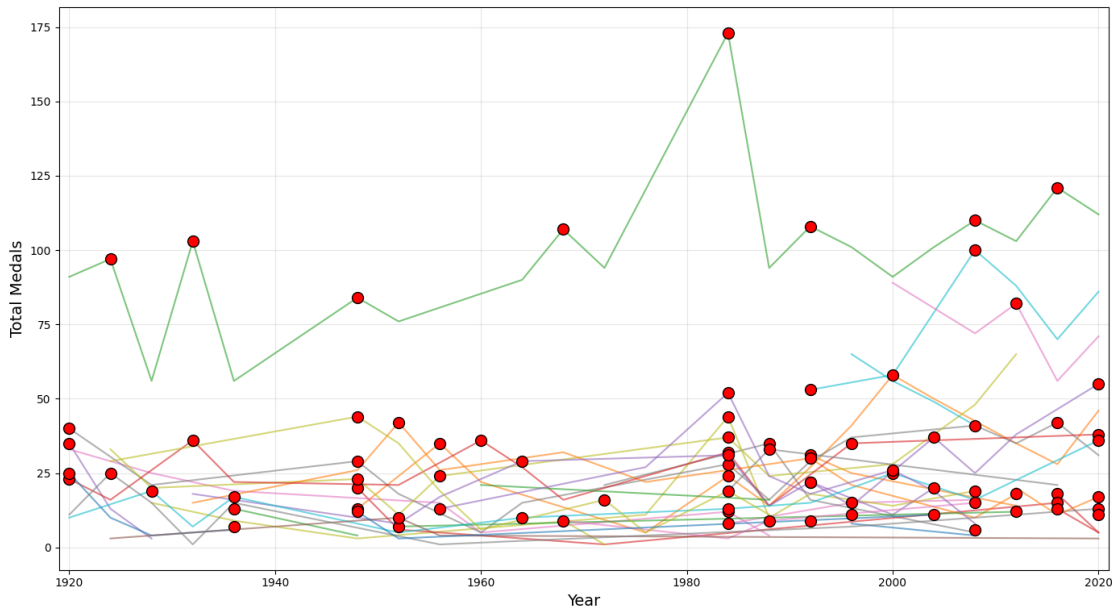

Figure 15: Filtered Medal Count Trends (Change > 5)

Further analysis suggests that while doping can temporarily boost medal performance, its long-term negative impact on a country's competitiveness in medal acquisition is difficult to reverse. These findings provide empirical support for anti-doping policies and highlight the trade-offs between short-term gains and long-term costs.

7 Evaluation of Strengths and Weaknesses

7.1 Strengths

This study introduces an Olympic medal prediction model based on graph data structures, which transforms traditional tabular data into ordered graph structures, effectively exploring the complex temporal and inter-country relationships in Olympic medal distributions. By segmenting data into blocks by country and into 35 Slices by time, this innovative data structure effectively showcases each country's comprehensive performance across different years. The model not only uncovers intrinsic connections between sports but also analyzes fundamental differences among countries in the same Olympic Games. The design of the graph data structure represents inter-country relationships through edges, providing richer information for subsequent analysis.

In addition, we augmented the data by extracting information such as ethnicity, geography, and veteran athletes in the data; some of this information is at the country level, and instead of putting it into slices, we built an additional set of *Info*. whereas features such as ethnicity and whether or not they are a veteran athlete are put into the node, which is able to better augment the subsequent prediction and learning.

Additionally, the model employs our improved LSTM model—TG-GRU, which effectively handles temporal trends in historical data, learning deep patterns from Olympic historical data to predict future medal distributions. Compared with the traditional recurrent neural network (RNN), on the one hand, it reduces the forgetting problem in the transmission process, and on the other hand, Compared to LSTM, it features a simpler structure and faster training speed. What' more, the graph representing

sports and the info representing geographic information are processed separately, which makes the interpretation better.

By comparing the performance of different models horizontally and validating data accuracy at various time points vertically, we demonstrate the strategy's advantage in prediction accuracy, proving its effectiveness and innovation in Olympic medal prediction.

7.2 Weaknesses

Despite the model's strong advantage in prediction accuracy, its "black box" nature may limit its application in scenarios requiring transparency and interpretability. Furthermore, due to the complex historical records of Olympic data, some countries' data may be missing or inconsistent, and the results of data cleaning used for model training can impact accuracy.

Nevertheless, the innovation of this study lies in proposing a new data processing and prediction method, providing valuable references for future Olympic medal predictions and offering a new modeling approach for similar multidimensional, multi-temporal scale problems.

8 Conclusion

In conclusion, this study has successfully developed a novel predictive model for Olympic medal outcomes, leveraging an enhanced ordered-graph data structure (Brick & Slide) and a Triple-Gated Gate Recurrent Unit (TG-GRU) algorithm. The model effectively integrates historical Olympic data and provides insightful predictions for the 2028 Olympic Games. Key findings include the identification of medal trends for leading nations, the impact of event diversity on medal outcomes, and the role of exceptional coaching in driving athletic performance. Furthermore, the study highlights challenges such as data gaps and the model's inherent complexity, while offering innovative approaches to address these issues.

The predictive model demonstrates significant advantages in precision and robustness compared to traditional methods, as evidenced by lower RMSE values and consistent performance across datasets. Beyond medal predictions, the study provides actionable insights into strategic areas such as anti-doping measures, breakthrough achievements for emerging nations, and the optimization of resource allocation for event participation. By balancing innovative methodologies with practical applications, this research not only advances the understanding of Olympic dynamics but also provides valuable tools for policymakers, coaches, and athletes aiming to optimize performance on the global stage.

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Appendices

How do we get edges? Here, we demonstrate code logic.

```
# Data Loading
Load 'all_blocks_dict.pkl'

# Model and Functions
MLP Model: Two fully connected layers (ReLU activation)
Contrastive Loss: Compute positive and negative sample loss using cosine similarity
Adjacency Matrix: Connect nodes if similarity > threshold

# Data Preprocessing
Extract samples and labels

# Model Initialization
Set input, hidden, and output dimensions
Initialize MLP and Adam optimizer

# Model Training
Repeat for 100 epochs:
  For each sample:
    Generate embeddings, compute loss, and update model
  Print average loss

# Adjacency Matrix Generation
For each sample:
  Compute embeddings and similarity matrix

# Update and Save
Attach adjacency matrices to data dictionary
Save as 'all_blocks_dict_with_arch.pkl'

```

AI Use Report

1.Open AI *ChatGPT*(Jan 26,2025 version)

Query1: Olympic Medals prediction‘s factor

Output: Predicting Olympic medals can be quite complex, as many factors contribute to a country's performance in the games. Here are some key factors that generally influence medal predictions:

Past Performance & Historical Data: Countries with a strong Olympic history (e.g., the United States, Russia, China) often perform well, but historical data alone isn’t always enough. It provides a baseline to gauge expectations.

Talent Pool & Athlete Development: Countries with well-established sports systems and strong youth programs tend to produce top athletes. The depth of talent in specific sports (e.g., swimming in the U.S., gymnastics in Russia) can also play a huge role.

Training & Infrastructure: Countries that invest heavily in sports infrastructure, training facilities, and coaches generally see better results. This includes things like sports science, nutrition, recovery strategies, and coaching expertise.

Funding & Sponsorship: Financial backing can provide the resources necessary for athletes to train and compete at the highest level. Sponsorship can also provide financial security, allowing athletes to focus solely on training.

Team/Individual Sports Focus: Some countries excel in team sports (e.g., basketball, soccer) while others dominate individual sports (e.g., athletics, gymnastics). The mix of sports in each Olympic cycle can affect predictions.

Mental Toughness & Psychological Factors: The mental aspect of the Olympics is huge. Athletes who can handle the pressure, cope with expectations, and perform under stress often outperform their peers.

Age of Athletes: The peak performance age varies by sport. In some events, younger athletes might perform better (e.g., gymnastics), while in others, experience and maturity are key (e.g., marathon running).

Changes in Regulations/Equipment: Innovations in sports equipment or changes in rules (e.g., swimwear advancements, scoring systems) can level the playing field or give certain countries/athletes a competitive advantage.

Home Advantage: Hosting countries tend to perform better due to familiarity with the environment, local support, and sometimes increased funding and resources allocated for the event.

Health and Injuries: Athletes’ health is a huge factor, and injuries can dramatically affect medal predictions. A key player or athlete being injured or out of form can change the outcome of many events.

Political and Geopolitical Factors: Sometimes political situations (boycotts, tensions) can impact participation and medal outcomes, as we've seen in past Olympics.

2.DeepSeek *deepseek* (Jan 26,2025 version, DeepSeek V3)

Ouery1:Write a Python program to convert the first level of the dictionary to a dictionary where the country is the key and the others are the value. The second level of the dictionary is the year as the key and the others are the value.

Output:Here is the Python program to convert the CSV data into a nested dictionary where the first-level keys are countries (NOC), and the second-level keys are years (Year). The values are the corresponding athlete data for each country and year. The input and output are in English.

Python Code:
```python
import csv
def csv_to_dict(file_path):
    data_dict = {}
    with open(file_path, mode='r', encoding='utf-8') as file:
        reader = csv.DictReader(file)
        for row in reader:
            noc = row['NOC']  # Country code
            year = row['Year']  # Year of participation

            # Initialize the nested dictionary if the country is not already a key
            if noc not in data_dict:
                data_dict[noc] = {}

            # Initialize the list for the year if the year is not already a key
            if year not in data_dict[noc]:
                data_dict[noc][year] = []
            # Append the athlete's data to the list for the corresponding year
            data_dict[noc][year].append(row)
    return data_dict
# File path
file_path = 'athletes_modified.csv'
# Convert CSV to dictionary
result = csv_to_dict(file_path)
# Print the result
for noc, years in result.items():
    print(f"Country: {noc}")
    for year, athletes in years.items():
        print(f"  Year: {year}")
        for athlete in athletes:
            print(f"    Athlete: {athlete['Name']}, Event: {athlete['Event']}, Medal: {athlete['No medal']}")
```

3, DeepL *DeepL* （Jan 26,2025 version）

Query1:Translate “Les Jeux olympiques ne sont pas de simples championnats du monde, mais un festival quadriennal de la jeunesse universelle célébré par chaque génération.”

Output:The Olympic Games are not just ordinary world championships but a quadrennial festival of universal youth celebrated by each succeeding generation.